Chapter 1 exercises

Use real column vectors. Unless stated otherwise, show the conditions and derivation, rather than only the final answer.

CH01-E1

Let $v = (2, 1)^T$ and $w = (1, 2)^T$. Find real c, d such that

$$cv + dw = (5, 4)^T.$$

State whether the solution is unique and explain why.

CH01-E2

For $u = (3, 4)^T$ and $v = (4, 3)^T$, answer the following.

1. Find the cosine of their angle.
2. Check the Cauchy–Schwarz inequality.
3. Find a nonzero z orthogonal to u, and describe the complete set of vectors orthogonal to u.

CH01-E3

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Compute $A(2, -1)^T$ by the column and row views. Check that they agree.

CH01-E4

Let $a_1 = (1, 2)^T$ and $a_2 = (2, 4)^T$.

1. Determine whether a_1, a_2 are independent.
2. Determine whether

$$a_1 x_1 + a_2 x_2 = (3, 5)^T$$

has a solution. 3. Replace the right-hand side by $(3, 6)^T$ and find the complete solution set.

CH01-E5

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Check invertibility and derive A^{-1}. Verify the inverse by multiplication rather than quoting an answer.

CH01-E6

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $x = (1, -1, 2)^T$. State input and output dimensions, then compute Ax by column combinations and row dot products, checking the results term by term.

CH01-E7

Extension; prerequisites: vector spaces, linear independence, and bases.

1. In $\mathcal{P}_2$, use the basis $B = (1, t, t^2)$ and let $p_1 = 1 + 2t$, $p_2 = 1 - t$, $p_3 = t + t^2$. Form the coordinate matrix D with these three coordinate columns. Determine independence and express $q(t) = 4 + 3t - t^2$ as a linear combination of p_1, p_2, p_3.
2. In $M_2(\mathbb{R})$, use column-major coordinates and let $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $A_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Determine whether the matrices are independent. If they are, determine whether they form a basis of $M_2(\mathbb{R})$.

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CH01-E8

Extension; prerequisites: eigenvalues, determinants, and polynomial factorization over the complex numbers.

Let

$$J = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}, \qquad r(t) = t^3 - 2t + 4.$$

1. Without assuming diagonalizability, prove for every $A \in \mathbb{C}^{n \times n}$ that $Av = \lambda v$ implies $r(A)v = r(\lambda)v$. Prove the reverse spectral inclusion by factoring a polynomial, and apply the result to J.
2. Use $J = 3I + N$, $N^2 = 0$, to compute $r(J)$ completely.
3. If another matrix B has eigenvalue 2 with a one-dimensional eigenspace, does having only one eigenvalue imply $B = 2I$? Give a counterexample or a justification.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 1.

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